Lecture 01

Math 4741: Mathematical Analysis

Chapter 1: Review of Probabilities

Texts:

1. Markov Processes for Stochastic Modeling – Oliver Ibe
2. Introduction to Probability and Stochastic Process – Goodman & Yates
3. Introduction to Probability Models – Sheldon Ross

Course Plan

Course Contents

- Review of Probability, Random variables
- Stochastic processes, Markov chains and simple queuing theory
- Applications to programs and algorithms analysis
- Computer systems performance and reliability modeling
- Renewal Theory, Distribution of N(t), Limit theorems and their applications, Renewal reward process,
- Semi-Markov process
- Techniques and models to develop and demonstrate wide range of problems associated with the design and analysis of various probabilistic systems in Computer Science.

Course Objectives

The subject aims to equip the students such that they will be able to:

1) Understand random processes and their behavior,
2) Select suitable random process to model the behavior of computer system,
3) Compute long-term probabilities or values of interesting parameters associated with applications that evolve over time, and
5) Make appropriate conclusions and forecasts

Course Outcomes and Program Outcomes

CO	Description	PO
CO1	Understand the knowledge of stochastic processes, and determine the steady-state Solutions of such processes associated with complex engineering problems	PO1
CO2	Apply knowledge of stochastic processes to model and analyze the performance of complex engineering problems	PO2
CO3	Apply knowledge of mathematics to interpret patterns and conclusions from complex computer problems	PO4

Weekly plan for course content and mapping with Cos

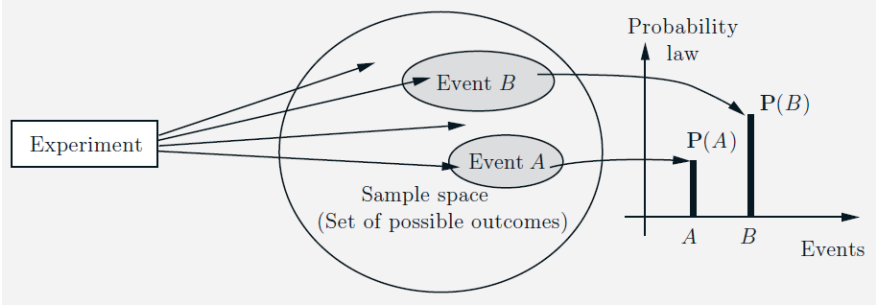
Weeks	Chapters/Topics	COs
1	Chapter 1: Review of Probability Chapter 2: Introduction to Stochastic Processes	CO1
2	Chapter 3: Counting Processes (Poisson Processes)	CO1
3	Chapter 3: Counting Processes (Poisson Processes and Bernoulli Processes)	CO1
	Quiz 1: 03/12/2025 – Poisson Process	
4	Chapter 4: Discrete-Time Markov Chains	CO1, CO2, CO3
5	Chapter 4: Discrete-time Markov Chains	
6	Chapter 5: Continuous-Time Markov Chains	CO1, CO2
	Quiz 2: 24/12/2025 - DTMC	
7	Chapter 5: Continuous-Time Markov Chains Chapter 6: Queueing Theory	CO2, CO3
8	Chapter 6: Queueing Theory	
9 – 10	Midterm Examination	

Chapter 1: Review of Probabilities

Probability

Probability Models

Mathematical description of a random experiment



<u>Sample Space</u> Set of all possible outcomes is said to be the sample space
The set can be a discrete set or a continuous set

<u>Event Space</u>
Collectively exhaustive and mutually exclusive set of events

Probability Axioms
<u>Axiom 1</u> (Non-negativity): If A is an event, then the probability of A satisfies

$$P[A] \geq 0$$

<u>Axiom 2</u> (Additivity): If A_1 and A_2 are two disjoint events, then

$$P[A_1 \cup A_2] = P[A_1] + P[A_2]$$

<u>Axiom 3</u> (Normalization): The probability of all elements in the sample space must be sum to 1

$$P[S] = P[\cup_{i=1}^{n} \omega_i] = \sum_{i=1}^{n} P[\omega_i] = 1$$

Simple Extension of the Axioms

1. If $A_1, A_2, \dots, A_n$ is a disjoint set, then
$$P[\cup_{i=1}^{n} A_i] = P[A_1] + P[A_2] + \cdots + P[A_n]$$

2. If $A \cap B \neq \phi$, A and B are not mutually exclusive, then
$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

3. $P[A^c] = 1 - P[A]$

4. If $A \subset B$, then $P[A] \leq P[B]$

Joint Probability

The probability of the intersection of two events, $P[A \cap B]$ or $P[A, B]$ or $P[AB]$

It is the simultaneous occurrence of two events

If events A and B are mutually exclusive, they two cannot occur together and the probability $P[AB] = 0$, because $A \cap B = \phi$

Conditional Probability:

- Partial information is given, which changes the sample space.
- The probabilities are calculated in the modified sample space usually with less number of elements.
- Conditional probability is dented by $P[A|B]$
- Probability of occurring event A, given that event B has already occurred

Probability of A, given B

$$P[A|B] = \frac{P[AB]}{P[B]}, P[B] > 0$$

Law of Multiplication (Product Rule)

$$P[AB] = P[A]P[B|A] = P[B]P[A|B]$$

Chain Rule

$$P[ABC] = P[A]P[B|A]P[C|AB]$$

$$P[A_1A_2 \dots A_n] = P[A_1]P[A_2|A_1]P[A_3|A_1A_2] \dots P[A_n|A_1A_2 \dots A_{n-1}]$$
$$= \prod_{i=1}^{n} P[A_i| \cap_{j=1}^{i-1} A_j]$$

Law of Total Probability (Sum Rule of Probability)

Let B be an event with $P[B] \geq 0$ and $P[B^c] \geq 0$, then for an event A

$$P[A] = P[A|B]P[B] + P[A|B^c]P[B^c]$$

Let $B = \{B_1, B_2, \dots, B_n$ be an event space, then for an event A

$$P[A] = P[A|B_1]P[B_1] + P[A|B_2]P[B_2] + \dots + P[A|B_n]P[B_n]$$

$$= \sum_{i=1}^{n} P[A|B_i]\, P[B_i]$$

Bayes Theorem

Let $B_1, B_2, \dots, B_n$ *be an event space of the sample space S of an experiment. If for* $i = 1, 2, \dots, n$, $P[B_i] > 0$ *then for any event* A *of* S *with* $P[A] > 0$,

$$P[B_i|A] = \frac{P[AB_i]}{P[A]}$$
$$= \frac{P[A|B_i]P[B_i]}{P[A|B_i]P[B_1]+\cdots+P[A|B_i]P[B_n]}$$

$B_1, B_2, \dots, B_n$ are called the hypothesis, $P[B_i]$ is called the **prior** probability of B_i, and the conditional probability $P[B_i|A]$ is called the **posterior** probability of B_i.

<u>Independence</u>

Knowing that B has occurred, does not change the probability of occurring event A

$$P[A|B] = P[A]\ , P[B] > 0$$

$$A \perp B \ \triangleq A \text{ is independed of } B$$

$$A \perp B = B \perp A$$

Independence is defined from the Law of multiplication

$$P[AB] = P[A|B]P[B] = P[A]P[B]$$

$$P[A \cap B] = P[A]P[B]$$

Two events are independent, if their joint probability is the product of their individual probabilities.

Random Variables

A RV is a single-valued real function, $X(\omega)$, that assigns a real number to each outcome $\omega \in S$.

Maps elements S onto points on the number line

X is used to represent a RV instead $X(\omega)$

The event E_x, where x is a real value, defines the subset of S that are pointed to x

$$E_x = \{\omega | S(\omega) = x\} = \{X = x\}$$

Distribution Functions

A distribution function represents a collection of probabilities

- Each probability is related to a real number, x
- Represents the chance of occurring the event represented by x

Type of Random Variables:

- <u>Discrete random variables</u>
  - Possible values are from a discrete set
  - Number of values are either finite or countably infinite
- <u>Continuous random variables</u>
  - Possible values are from an interval
  - Number of values are uncountable
- <u>Mixed random variables</u>

<u>Events associated with a random variable are</u>:

- The random variable has a specific value
  - $\{X = x\}$ or more specifically $\{X(\omega) = x\}$
- The random variable has a value which is less than or equal to a specific value:
  - $\{X \leq x\}$ or $X(\omega) \leq x\}$
- The random variable has value which is greater than a specific value
  - $\{X \geq x\}$ or $\{X(\omega) \geq x\}$

<u>Possible distribution functions are</u>:

- Probability Mass function (PMF): The probability that a RV X has a specific value x
  - $P_X(x) = P[X = x]$

- Cumulative distribution function (CDF): The probability that a RV has a value which is less than or equal to a specific value x
  - $F_X(x) = P[X \leq x]$

- Complementary cumulative distribution function (CCDF): The probability that a RV has a value which is greater than a specific value x
  - $P[X > x]$
- Probability density function (PDF): The amount of probability per unit length for a continuous RV
  - $f_X(x) = \frac{d}{dx} F_X(x)$

Properties of Distribution Functions

1. $P_X(x) \geq 0$, $f_X(x) \geq 0$, $F_X(x) \geq 0$
2. $\sum_x P_X(x) = 1$, $\int_{-\infty}^{\infty} f_X(x)dx = 1$
3. $F_X(x)$ is non-decreasing, and it is continuous for a continuous RV
   $F_X(x) = \int_{-\infty}^{x} f_X(x)dx$

Expected Value

$$E[X] = \sum_x x\, P_X(x) = \int_{-\infty}^{\infty} x f_X(x) dx$$

Moments

$$E[X^n] = \sum_x x^n P_X(x) = \int_{-\infty}^{\infty} x^n f_X(x) dx$$

Central Moments

$$E[(X - \mu_x)^n] = \sum_x (x - \mu_x)^n P_X(x) = \int_{-\infty}^{\infty} (x - \mu_x)^n f_X(x) dx$$

Bivariate Random Variables or Bivariate Distribution

Simultaneous occurrence of a pair of values associated with 2 RVs defined on the same sample space

Assign a point on the $x - y$ plane as the outcome of an experiment

$$P_{XY}(x, y) = P[X = x, Y = y]$$
$$F_{XY}(x, y) = P[X \leq x, Y \leq y]$$
$$f_{XY}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x, y)$$
$$F_{XY}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{XY}(x, y) dx dy$$

Marginal Distribution

$$P_X(x) = \sum_y P_{XY}(x, y)$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)dy$$

Covariance and Correlation Coefficient

$$C_{XY} = Cov[X, y] = \sigma_{XY} = E[(x - \mu_X)(y - \mu_Y)]$$

$$= E[XY] - \mu_X \mu_Y$$

$$\rho_{XY} = \frac{C_{XY}}{\sigma_X \sigma_Y}$$

$-1 \leq \rho \leq 1$

Conditional Distribution

$$P_{X|A}(x) = \frac{P_X(x)}{P[A]}, x \in A$$

$$P_{X|Y}(x|y) = \frac{P_{XY}(x,y)}{P_Y(y)}$$

$$f_{X|A}(x) = \frac{f_X(x)}{P[A]}, x \in A$$

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_{X(x)}}$$

Product Rule

$$P_{XY}(x,y) = P_{X|Y}(x|y)P_Y(y) = P_{Y|X}(y|x)P_X(x)$$

$$f_{XY}(x,y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x)$$

Sum Rule

$$P_X(x) = \sum_y P_{X|Y}(x|y)P_Y(y)$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y)dy$$

Conditional Expectation

$$E[X|A] = \sum_x xP_{X|A}(x) = \int_{-\infty}^{\infty} xf_{X|A}(x)dx$$

$$E[X|Y = y] = \sum_x xP_{X|Y}(x|y) = \int_{-\infty}^{\infty} xf_{X|Y}(x|y)dx$$

$$E[X] = \sum_y E[X|Y = y]P_Y(y) = \int_{-\infty}^{\infty} E[X|Y = y]f_Y(y)dy$$

Law of Total Expectation

Independence

Two RVs X and Y are independent if, for any subsets A and B of real numbers, the events $\{X \in A\}$ and $\{X \in B\}$ are independent,

$$P[X \in A, Y \in B] = P[X \in A]P[Y \in B]\backslash$$

Which can be easily extended, for any two real numbers x and y, to

$$P[X \leq x, Y \leq y] = P[X \leq x]P[Y \leq y]$$

If X and Y are two RVs defined on the same sample space, then X and Y are independent iif, for all real numbers x and y,

$$F_{XY}(x, y) = F_X(x)F_Y(y)$$

And is applicable for both discrete and continuous RVs

If

$$P_{X|Y}(x|y) = P_X(x)$$

Then, knowing the value of RV $Y = y$, does not change the probability $X = x$

$$P_{XY}(x, y) = P_{X|Y}(x|y)P_Y(y) = P_X(x)P_Y(y)$$

Similarly,

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

Multivariate Distribution

$$P_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = P[X_1 = x_1,\ldots,X_n = x_n].$$

$$\begin{aligned}
P_{X_i}(x_i) &= P[X_i = x_i] \\
&= P[X_i = x_i, x_j \in S_{X_j}], \quad i \le j \le n, i \ne j \\
&= \sum_{x_1 \in S_{X_1}} \cdots \sum_{x_{i-1} \in S_{X_{i-1}}} \sum_{x_{i+1} \in S_{X_{i+1}}} \cdots \sum_{x_n \in S_{X_n}} P_{X_1,\ldots,X_n}(x_1,\ldots,x_n), \qquad 1 \le i \le n
\end{aligned}$$

$$F_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = P[X_1 \leq x_1,\ldots,X_n \leq x_n]$$

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \frac{\partial^n F_{X_1,\ldots,X_n}(x_1,\ldots,x_n)}{\partial x_1,\ldots,\partial x_n}$$

$$f_{X_1}(x_1) = \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1} f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) dx_2 \ldots dx_n$$

Well-known Random Variables

Bernoulli Distribution

$$P_X(x) = \begin{cases} 1-p, & \text{for } x = 0, \\ p, & \text{for } x = 1, \\ 0, & otherwise. \end{cases}$$

Geometric Distribution

$$P_X(x) = \begin{cases} p(1-p)^{x-1}, & \text{for } x \geq 1; \\ 0, & otherwise. \end{cases}$$

Binomial Distribution

$$P_X(k) = \begin{cases} \binom{n}{k} p^k q^{n-k}, & \text{for } k = 0, 1, 2, \ldots; \\ 0, & otherwise. \end{cases}$$

Pascal Distribution

$$P_X(n) = \binom{n-1}{k-1} p^k q^{n-k}, \qquad n = k, k+1, \ldots.$$

Hypergeometric Distribution

$$P_X(x) = \begin{cases} \frac{\binom{r}{x}\binom{g}{n-x}}{\binom{r+g}{n}}, & x = 0, 1, \ldots, \min(r, n) \\ 0, & \text{otherwise.} \end{cases}$$

Poisson Distribution

$$P_X(x) = \begin{cases} \frac{(\lambda T)^x e^{-(\lambda T)}}{x!}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Discrete Uniform Distribution

$$P_X(x) = \begin{cases} \frac{1}{m-l+1}, & \text{for } x = l, l+1, l+2, \ldots, m; \\ 0, & otherwise. \end{cases}$$

Uniform Continuous Random Variables

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & \text{for } a \leq X \leq b; \\ 0, & \text{otherwise.} \end{cases}$$

Exponential Random Variables

$$f_X(x) = \lambda e^{-\lambda x}, x \geq 0$$

m-Erlang or Gamma Random Variables

$$f(x) = \begin{cases} \lambda e^{-\lambda x} \frac{(\lambda x)^{n-1}}{(n-1)!} & \text{if } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$f(x) = \begin{cases} \frac{\lambda e^{-\lambda x} (\lambda x)^{r-1}}{\Gamma(r)} & \textit{if } x \geq 0 \\ 0 & \textit{elsewhere} \end{cases}$$

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$$

Beta Random Variables

$$f(x) = \begin{cases} \dfrac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1} & \text{if } 0 < x < 1 \\ 0 & \text{otherwise,} \end{cases}$$

$$B(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx.$$

$$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

Normal Random Variables

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad \text{for } -\infty < x < \infty$$

Log Normal Random Variables

If $log(X)$ has the normal distribution with mean μ and variance σ^2, we say that X has the *lognormal distribution* with parameters μ and σ^2.

Multivariate Hypergeometric Distribution

$$P_{X_1,\ldots,X_r}(x_1,\ldots,x_r) = \frac{\binom{n_1}{x_1}\binom{n_2}{x_2}\cdots\binom{n_r}{x_r}}{\binom{n_1+n_2+\cdots+n_r}{n}}$$

Multinomial distribution

$$f(\boldsymbol{x}|n,\boldsymbol{p}) = \Pr(\boldsymbol{X}=\boldsymbol{x}) = \Pr(X_1 = x_1,\ldots,X_k = x_k)$$
$$= \begin{cases} \binom{n}{x_1,\ldots,x_k} p_1^{x_1}\cdots p_k^{x_k} & \text{if } x_1+\cdots+x_k = n, \\ 0 & \text{otherwise.} \end{cases}$$

Random Vectors

$$\begin{aligned} F_{\mathbf{X}}(\mathbf{x}) &= P[\mathbf{X} \leq \mathbf{x}] \\ &= F_{X_1, X_2, \ldots, X_n}(x_1, x_2, \ldots, x_n) \\ &= P[X_1 \leq x_1, X_2 \leq x_2, \ldots, X_n \leq x_n] \end{aligned}$$

$$\begin{aligned} P_{\mathbf{X}}(\mathbf{x}) &= P[\mathbf{X} = \mathbf{x}] \\ &= P_{X_1, X_2, \ldots, X_n}(x_1, x_2, \ldots, x_n) \\ &= P[X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n] \end{aligned}$$

$$f_{\mathbf{X}}(\mathbf{x}) = f_{X_1, X_2, \ldots, X_n}(x_1, x_2, \ldots, x_n).$$

Expected value vector

$$E[\mathbf{X}] = \mu_{\mathbf{X}} = \begin{bmatrix} E[X_1] & E[X_2] & \cdots & E[X_n] \end{bmatrix}^T$$

Covariance Matrix

$$Cov[\mathbf{X}] = \begin{bmatrix} \mathrm{Var}[X] & Cov[X_1, X_2] & \cdots & Cov[X_1, X_n] \\ Cov[X_2, X_1] & Var[X_2, X_2] & \cdots & Cov[X_2, X_n] \\ \vdots & \vdots & \ddots & \vdots \\ Cov[X_n, X_1] & Cov[X_n, X_2] & \cdots & Var[X_n, X_n] \end{bmatrix}$$

Correlation Matrix

$$R_{\mathbf{X}} = \begin{bmatrix} E[X_1^2] & E[X_1, X_2] & \cdots & E[X_1, X_n] \\ E[X_2, X_1] & E[X_2^2] & \cdots & E[X_2, X_n] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_n, X_1] & E[X_n, X_2] & \cdots & E[X_n^2] \end{bmatrix}$$

$$Cov[\mathbf{X}] = R_{\mathbf{X}} - \mu_{\mathbf{X}} \mu_{\mathbf{X}}^T.$$

Cross-Correlation and Cross-Covariance

$$\mathbf{R}_{\mathbf{XY}} = E[\mathbf{X}\mathbf{Y}^T].$$

$$\mathbf{Cov}[\mathbf{X}, \mathbf{Y}] = E[(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})(\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})^T].$$

Joint Gaussian Distribution

Random variables X and Y are jointly Gaussian if their joint PDF is given by

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} e^{-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2\left(1-\rho_{XY}^2\right)}}$$

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^2|\boldsymbol{C_X}|}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_X})^T \boldsymbol{C_X}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_X})}$$

Multivariate Joint Gaussian PDF of random vector X is given by

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{N/2}|\boldsymbol{C_X}|^{1/2}} e^{-(\boldsymbol{x}-\boldsymbol{\mu_X})^T \boldsymbol{C_X^{-1}} (\boldsymbol{x}-\boldsymbol{\mu_X})}$$

$$\Delta^2 = (\boldsymbol{x} - \boldsymbol{\mu_X})^T \boldsymbol{C_X^{-1}} (\boldsymbol{x} - \boldsymbol{\mu_X}).$$

The quantity Δ is called the Mahalanobis distance.